

Course

« *Computer Vision* »

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Computer vision – different levels

1. Low (Image processing)

- Image restoration
- Contrast enhancement
- Noise reduction

image → image

2. Medium

- Image segmentation
- Shape recognition

image → Features/attributes/
descriptors

3. High

- Scene understanding

image → Concepts

X. The SIFT: Introduction

- **Challenge:** Recognize the query inspite of variations like rotation, scale, illumination, occlusion.

Template/Query



Target (rich) image



⇒ **Solution:** Find and match interest points and features.

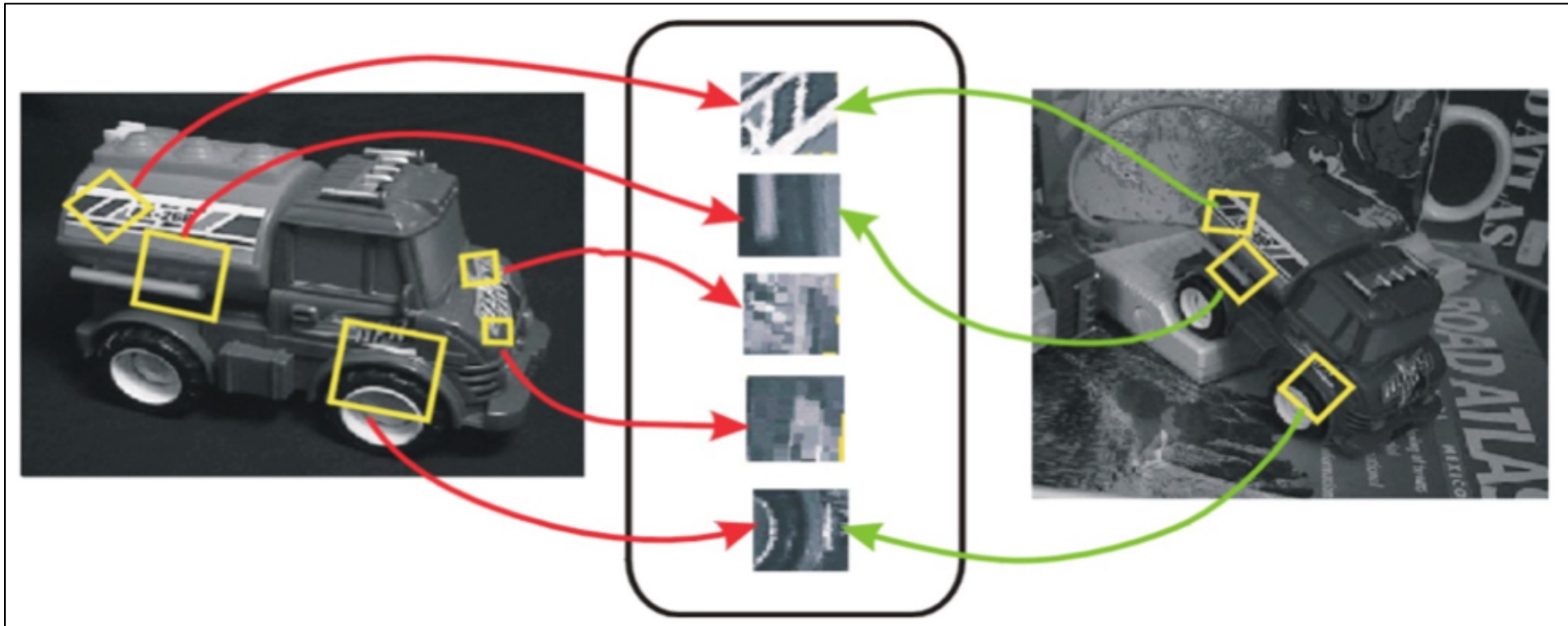
X. The SIFT: Introduction

- **Ref.:** David G. Lowe, **Distinctive image features from scale-invariant keypoints**, *International Journal of Computer Vision*, 60, 2 (2004). Cité 78589 fois
- It changed the way we approach many computer vision problems! (Including object recognition).
- Invariances:
 - Scaling
 - Rotation
 - Illumination

While still providing very good localization.

X. The SIFT: Introduction

- **General idea:** Image content is transformed into local feature coordinates that are invariant to translation, rotation, scale, and other imaging parameters



X. The SIFT: Introduction

Advantages:

- **Locality:** features are local, so robust to occlusion and clutter (no prior segmentation).
- **Distinctiveness:** individual features can be matched to a large database of objects.
- **Quantity:** many features can be generated for even small objects.
- **Efficiency:** close to real-time performance.
- **Extensibility:** can easily be extended to wide range of differing feature types, with each adding robustness.

X. The SIFT: Introduction

It consists of two parts:

1. A detector of interest area: keypoint localization
2. A local signature computation method.

Signature = descriptor = feature vector.

- SIFT Detector
- SIFT Descriptor
- Descriptor matching
- Applications making use of SIFT descriptors.

XI. The SIFT Detector

What really is an interest point in an image?



XI. The SIFT Detector

What really is an interest point in an image?

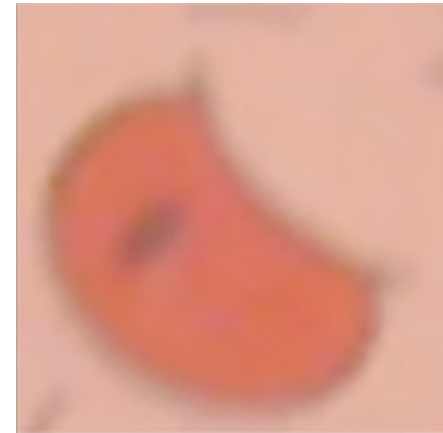


Different size, orientation, scale, lightening, brightness...

XI. The SIFT Detector

An interest point could be able to cope with variations

⇒ Identify points (representing the same visual content) in two images even if images are different.



XI. The SIFT Detector

An interest point could be able to cope with variations

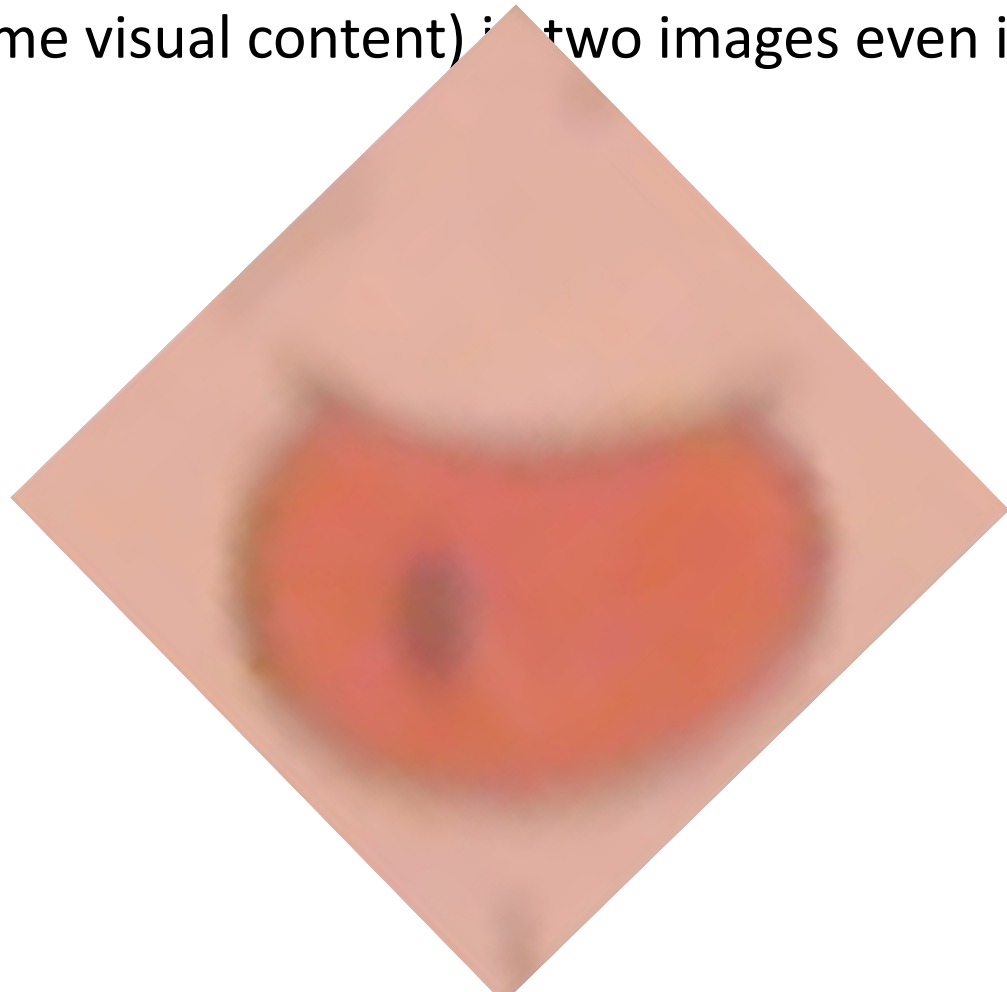
⇒ Identify points (representing the same visual content) in two images even if images are different.



XI. The SIFT Detector

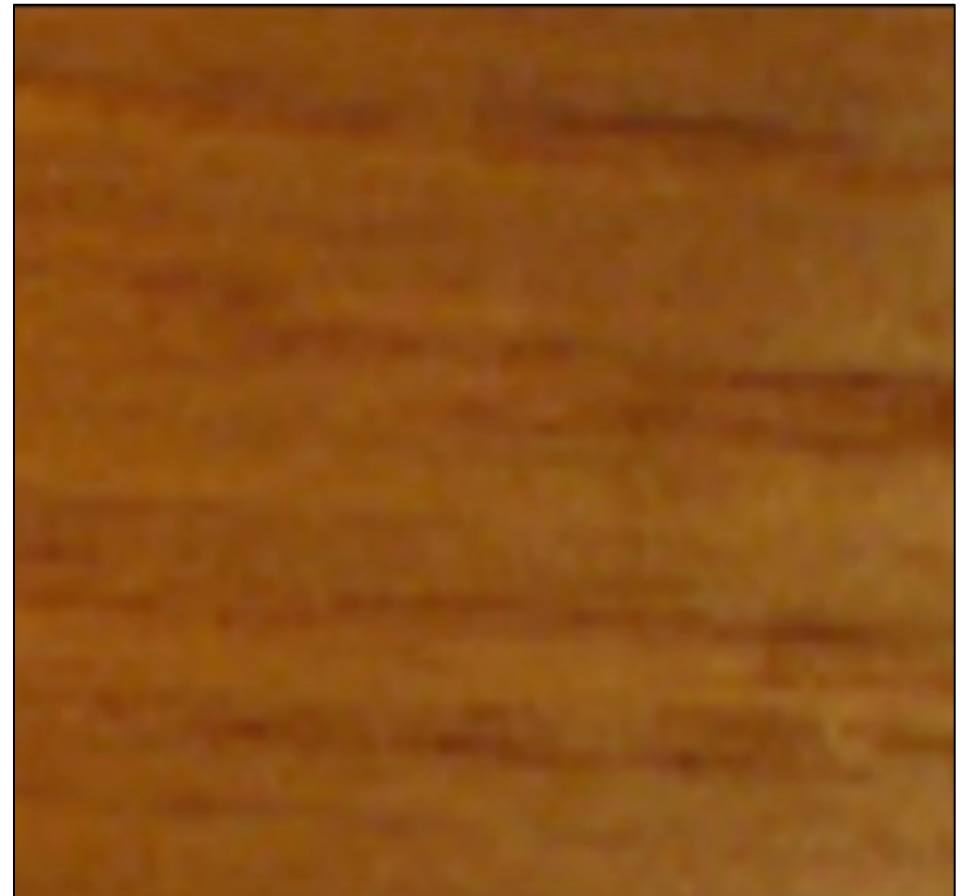
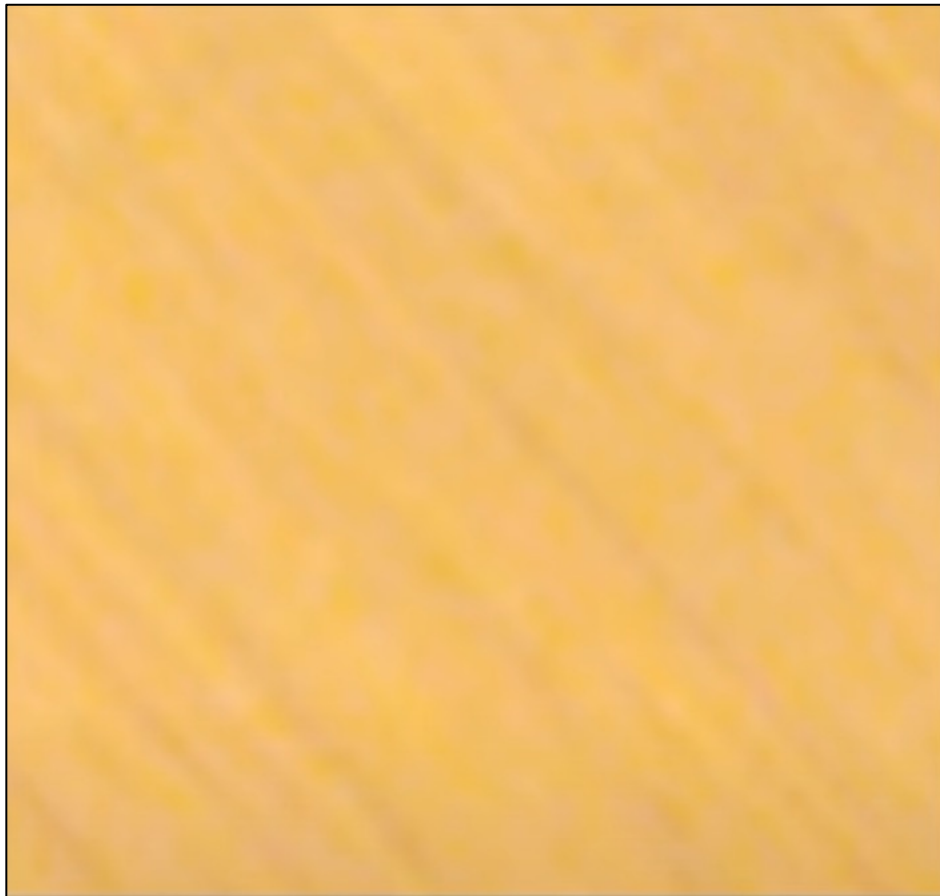
An interest point could be able to cope with variations

⇒ Identify points (representing the same visual content) in two images even if images are different.



XI. The SIFT Detector

Some patches/points in the image are not “interesting” (for image matching).



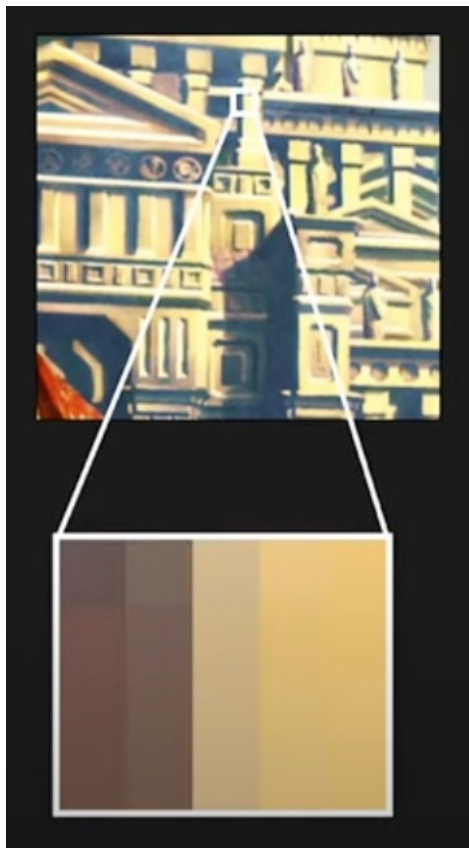
XI. The SIFT Detector

What are the desired properties of interest points:

- **Rich image content** within the window: brightness variations, color variations, etc.
- Has a **well-defined representation** (description/signature) for comparing/matching with other points.
- Has a **well-defined position** in the image.
- Should be **invariant to image rotation and scaling**.
- Should be **insensitive to lighting** changes.

XI. The SIFT Detector

Do lines/edges good interest points?



XI. The SIFT Detector

What about blobs?

- In computer vision, a **blob** refers to a region in an image that is **visually distinct** from its surroundings, typically in terms of **brightness, color, or texture**.
- Blobs can represent important parts of objects, corners, or structures like dots, spots, or junctions.
- In other terms, blobs are **areas where the image intensity remains relatively constant** but changes significantly at the boundaries.

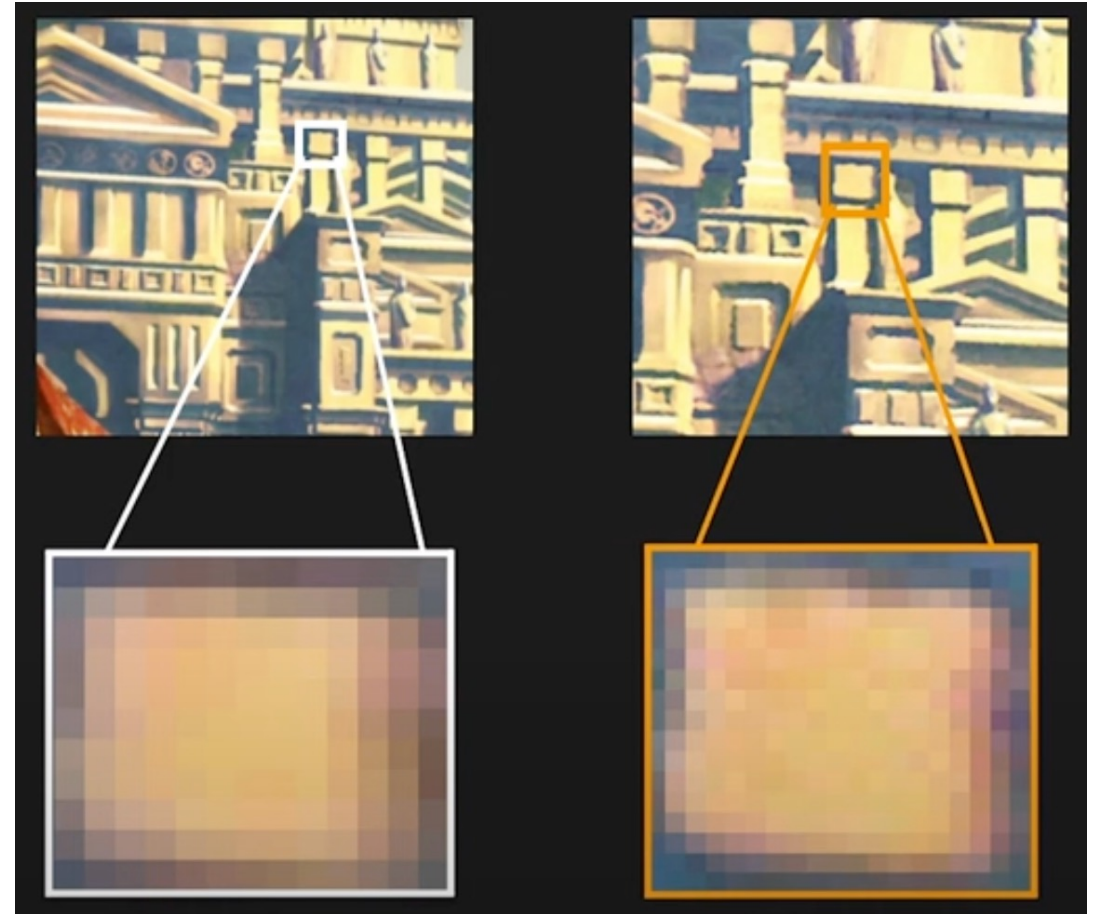
Characteristics of Blobs

- **Localized** in both space and scale.
- **Not necessarily circular**, though often approximated as such for analysis.
- **Scale-dependent/sensitive**: The same feature might look different at different scales.

XI. The SIFT Detector

What about blobs?

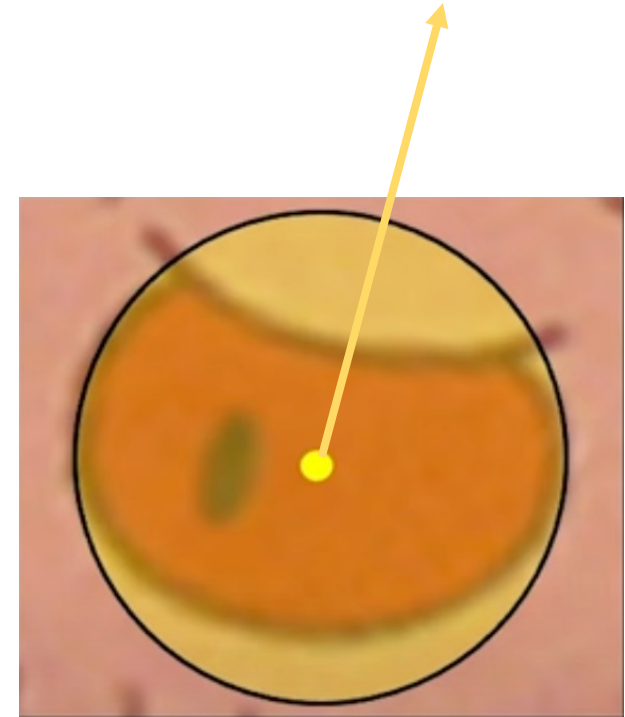
- Fixed position.
 - Consistent shape/appearance.
 - Definite size.
- ⇒ May be good as interest regions.



XI. The SIFT Detector

To build a local descriptor based on blobs, we need to:

- Locate the blob.
- Determine the size.
- Determine the orientation.
- Formulate an invariant description.

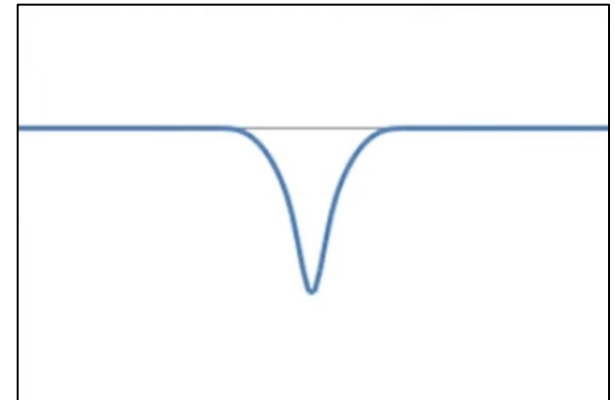
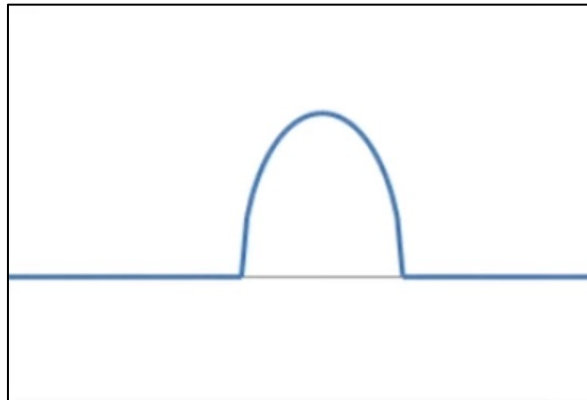
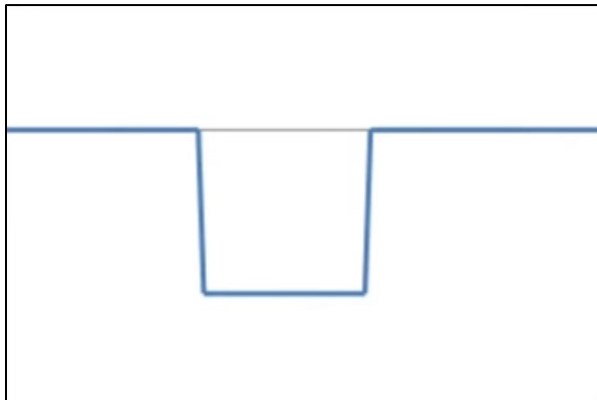
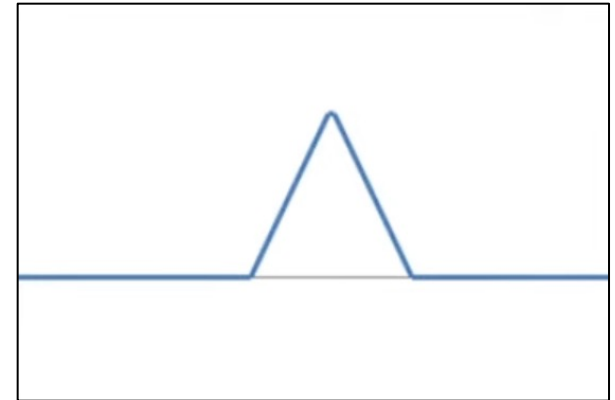
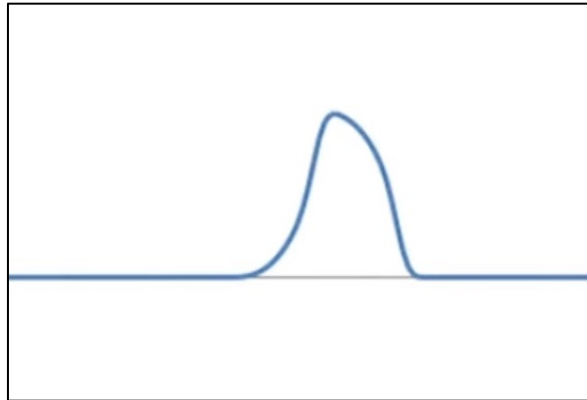
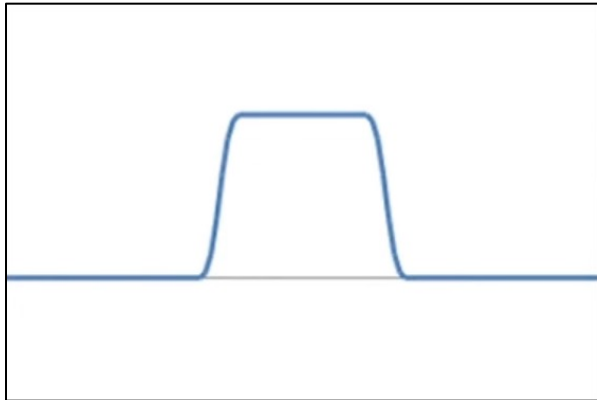


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XI. The SIFT Detector

Principle of blob detection – Let's start with 1D signals

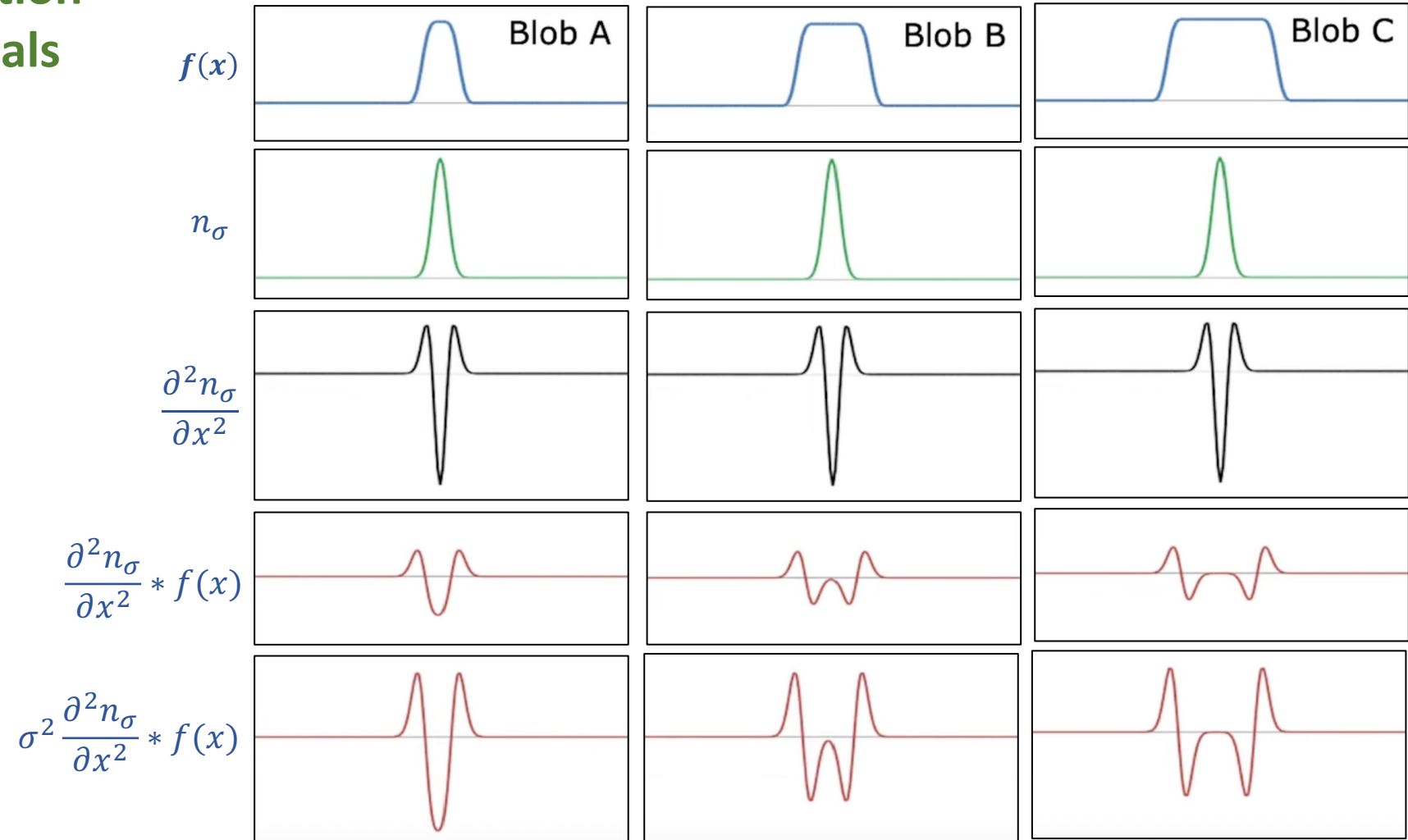
Examples of 1D blob structures



XI. The SIFT Detector

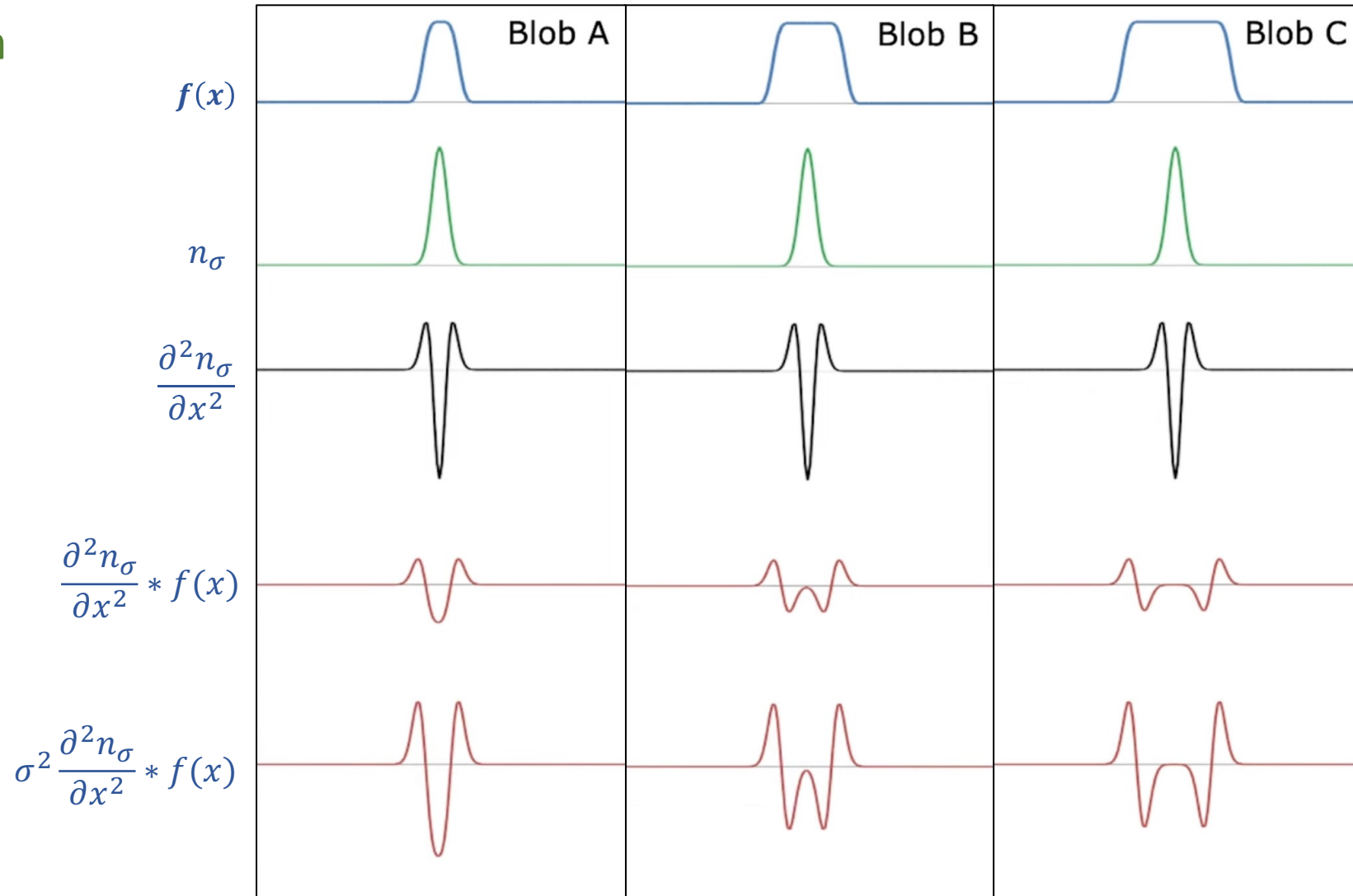
Principle of blob detection

Let's start with 1D signals



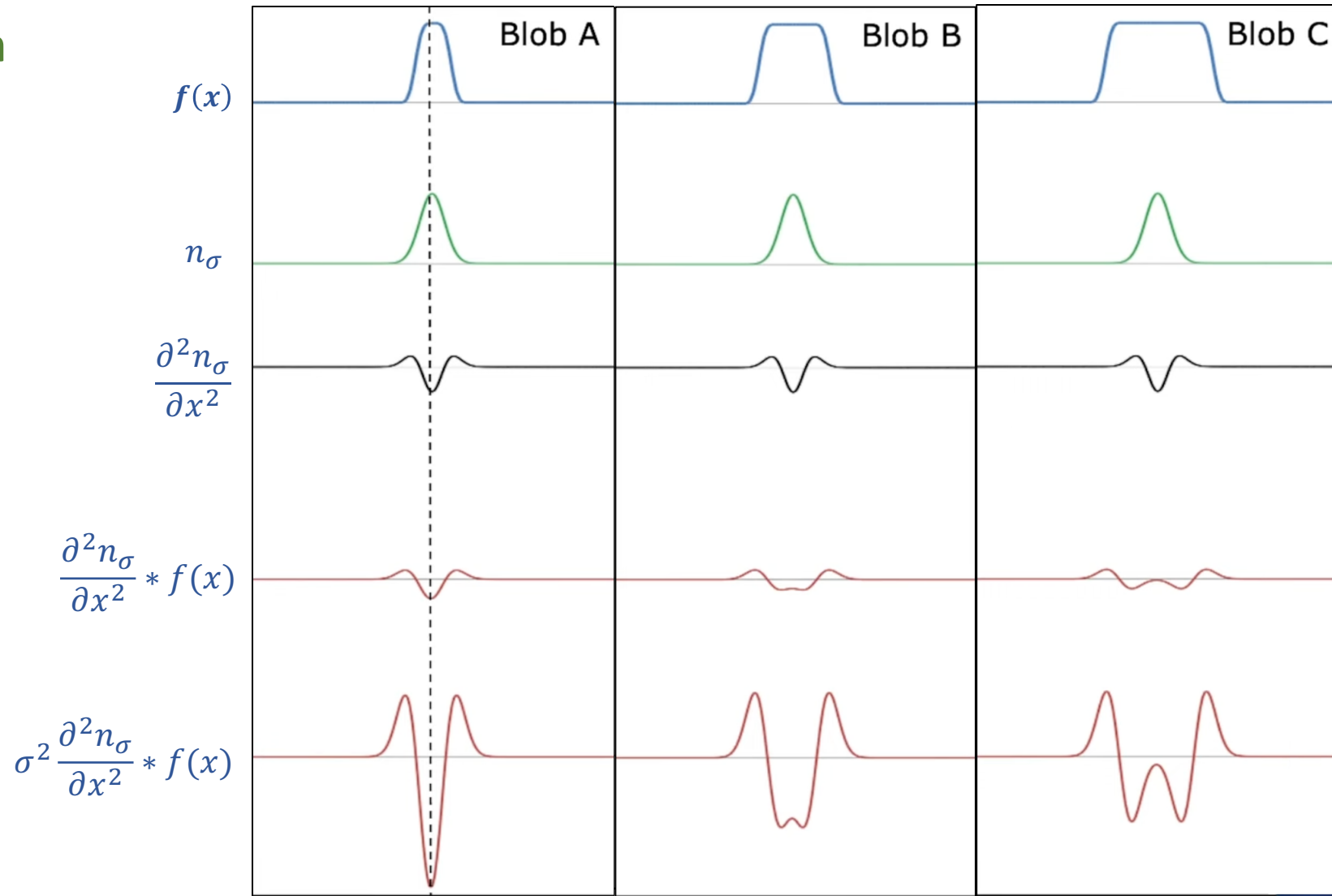
XI. The SIFT Detector

Principle of blob detection
Let's start with 1D signals



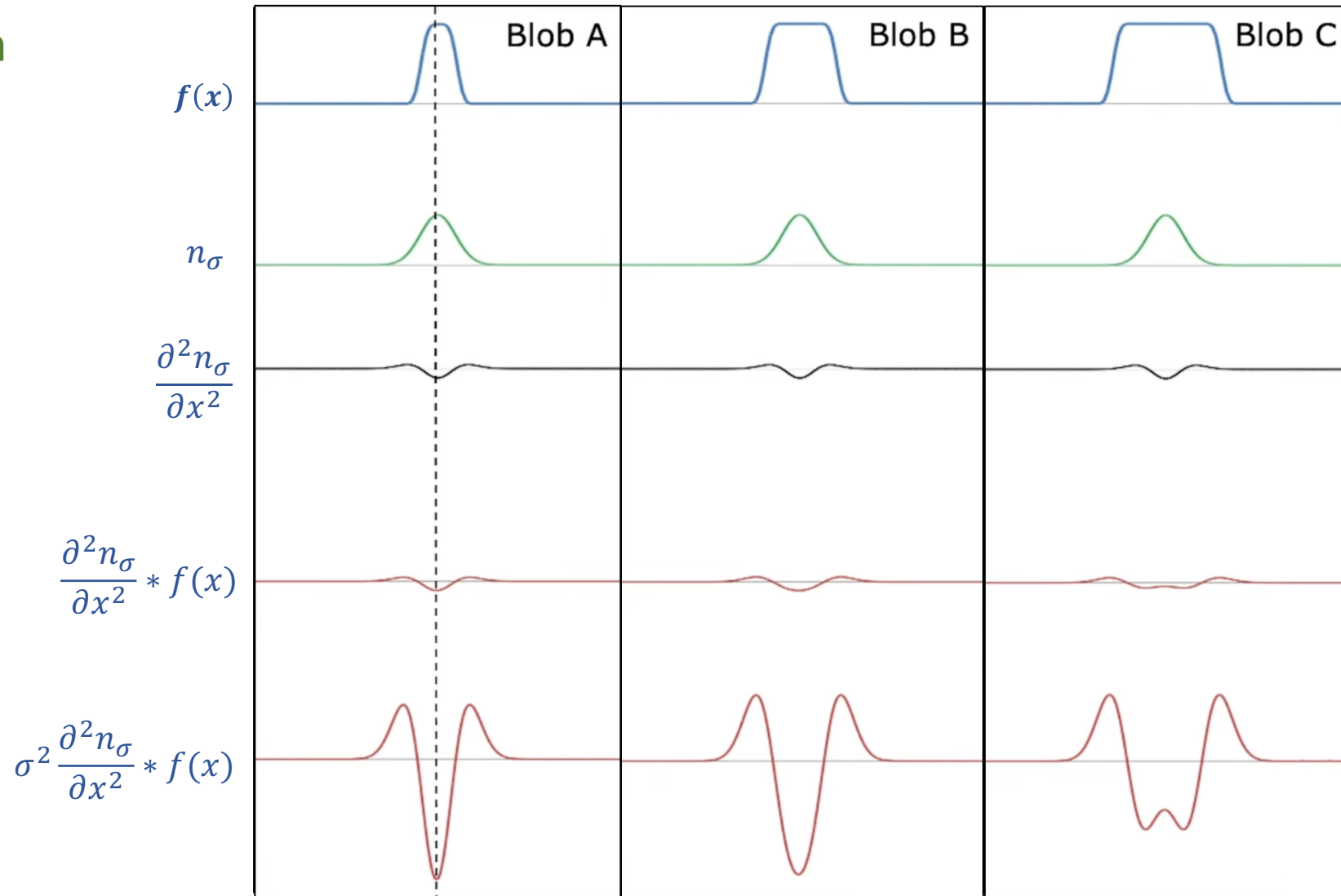
XI. The SIFT Detector

Principle of blob detection
Let's start with 1D signals



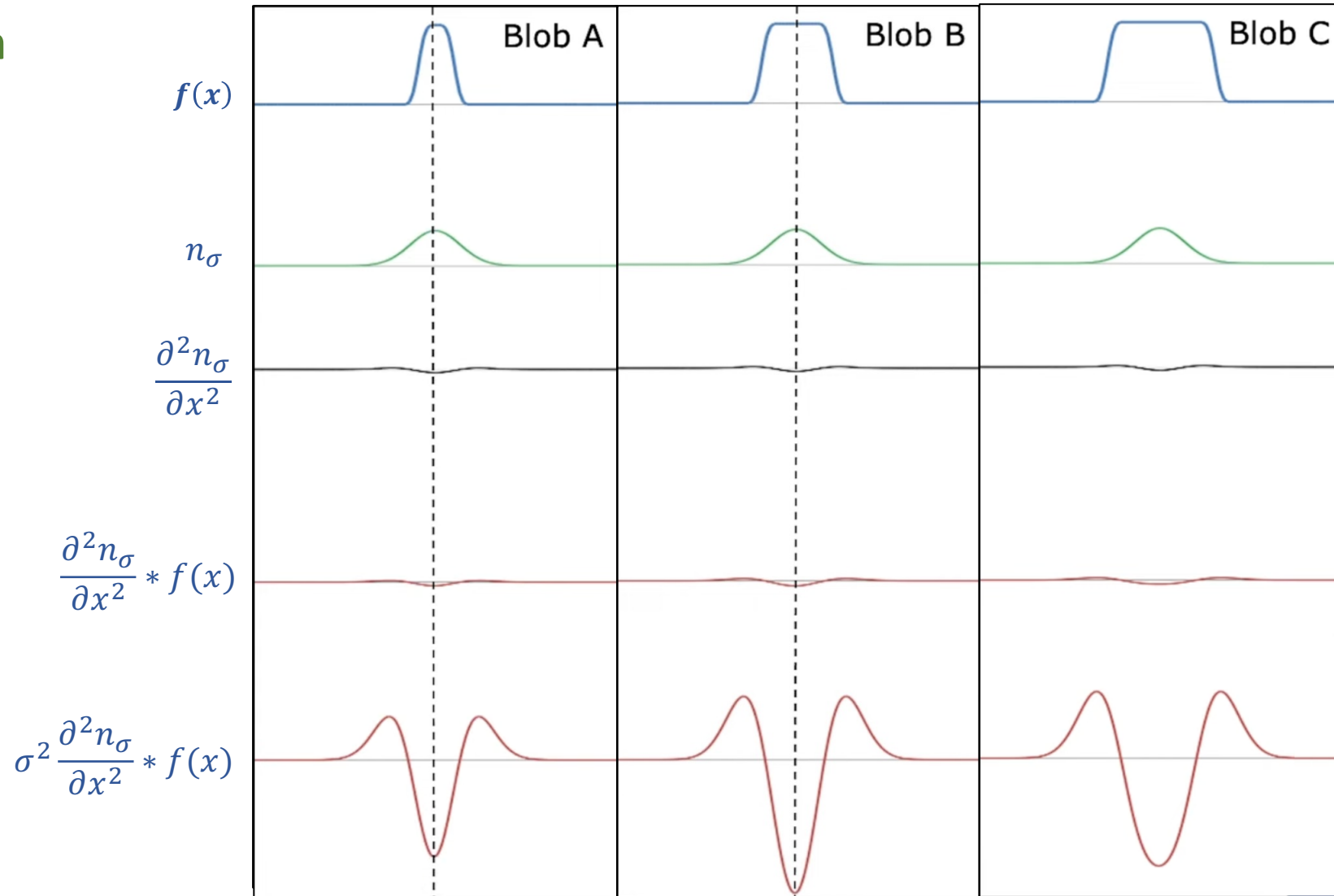
XI. The SIFT Detector

Principle of blob detection
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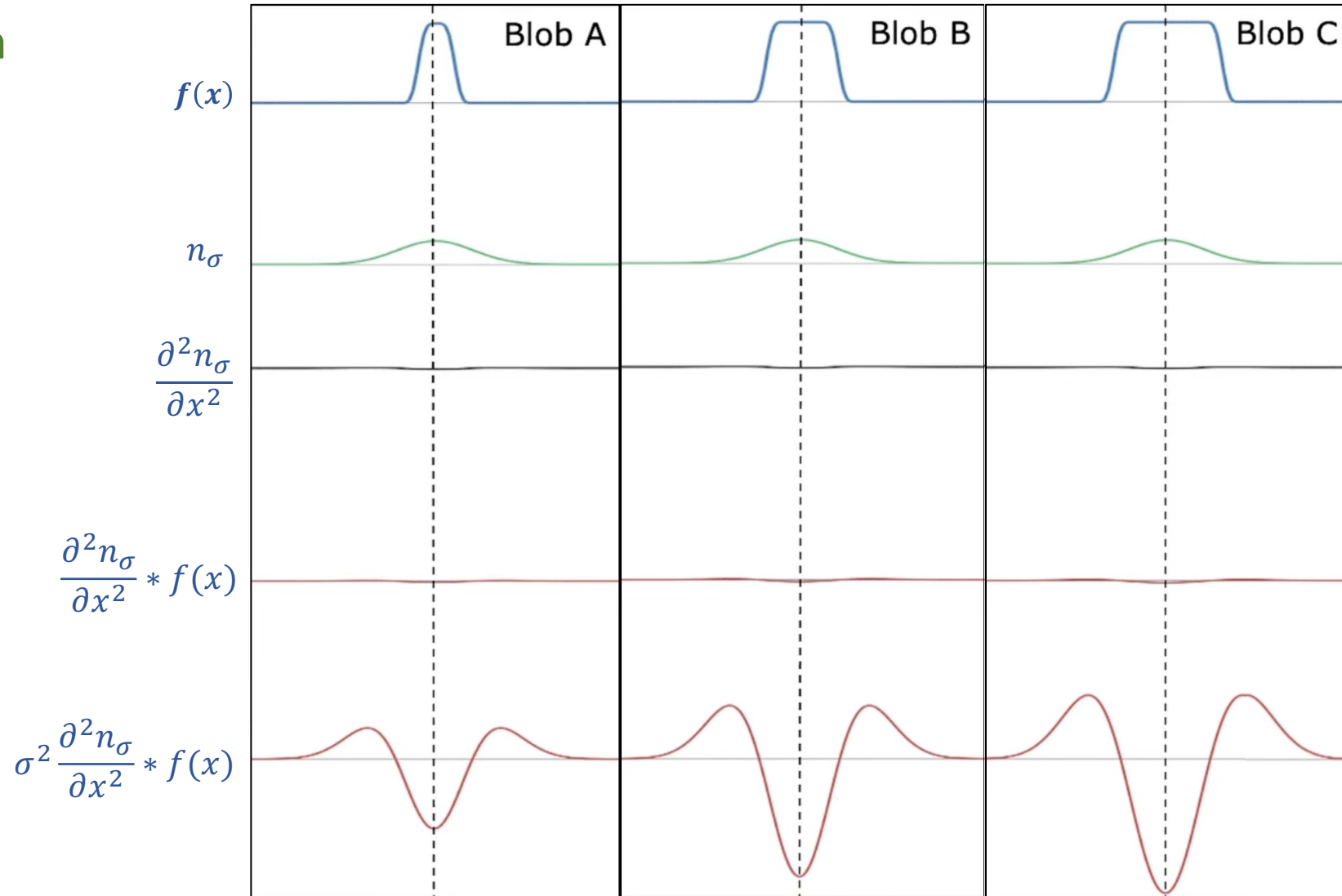
XI. The SIFT Detector

Principle of blob detection
Let's start with 1D signals



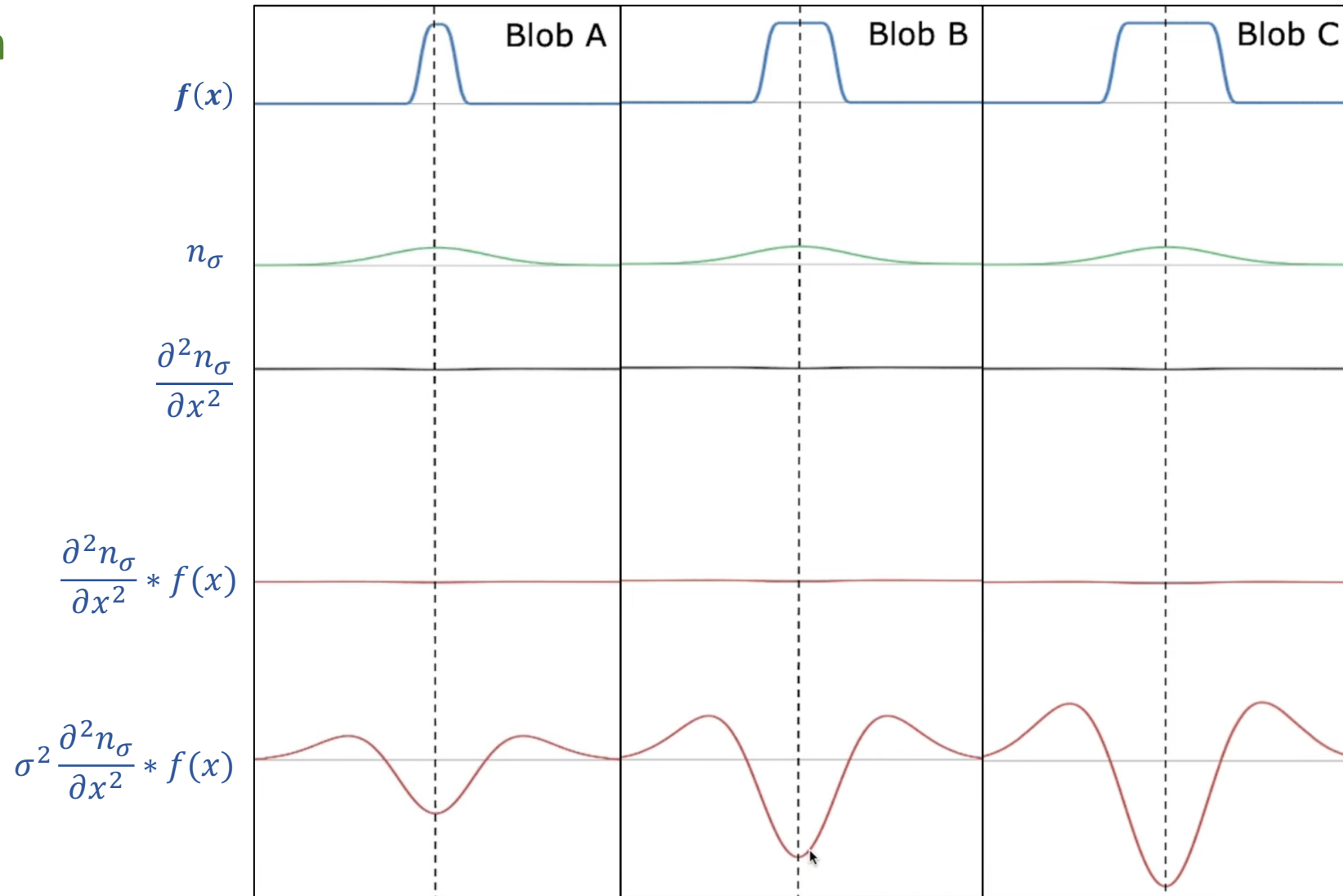
XI. The SIFT Detector

Principle of blob detection
Let's start with 1D signals



XI. The SIFT Detector

Principle of blob detection
Let's start with 1D signals



XI. The SIFT Detector

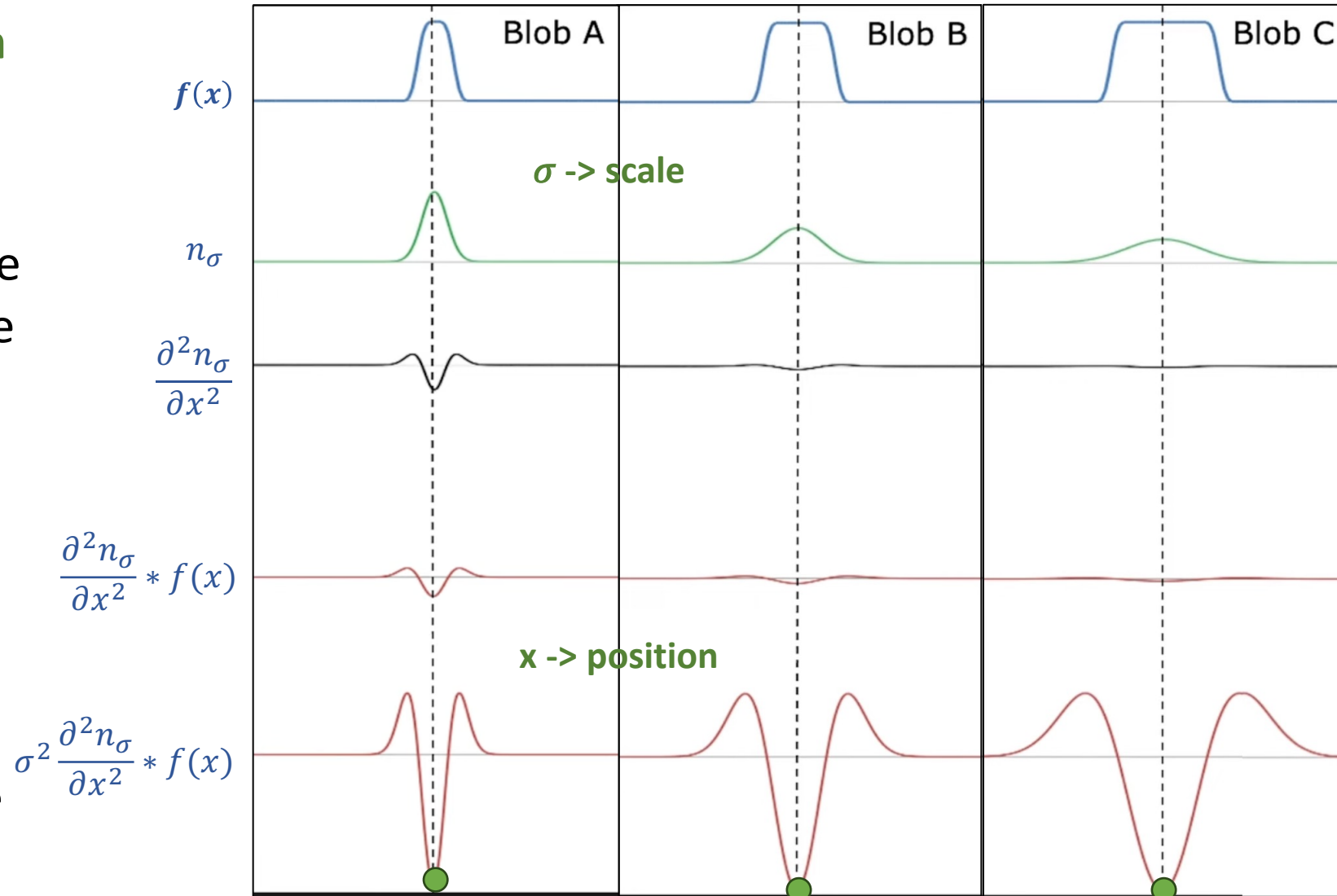
Principle of blob detection Let's start with 1D signals

Each blob is detected as a **minimum/maximum** of the σ -normalized 2nd derivative of the signal.

Each blob is detected at a specific value of σ

⇒ **The scale**

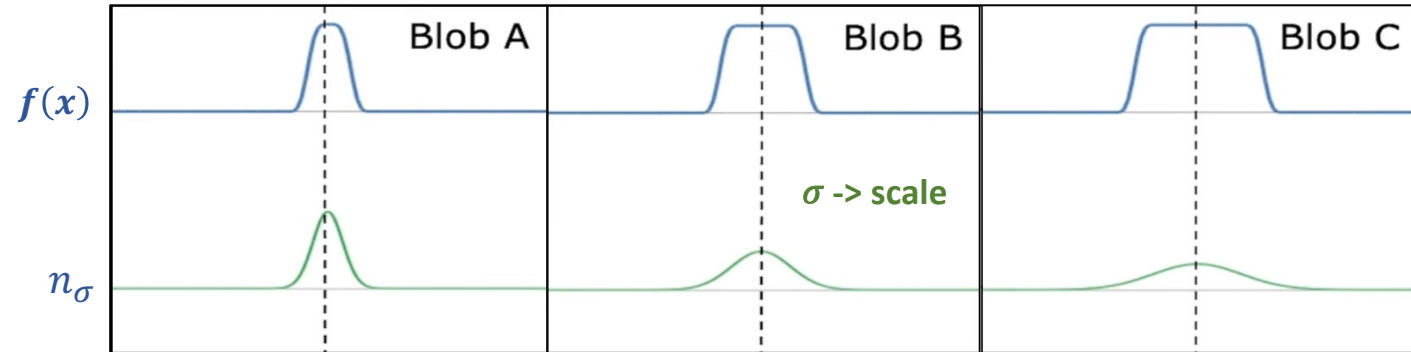
A blob is a minimum/maximum in the (x, σ) space.



XI. The SIFT Detector

Principle of blob detection

Let's start with 1D signals



⇒ The characteristic scale:

The σ at which the σ -normalized 2nd derivative of the signal attains its minimum/maximum.

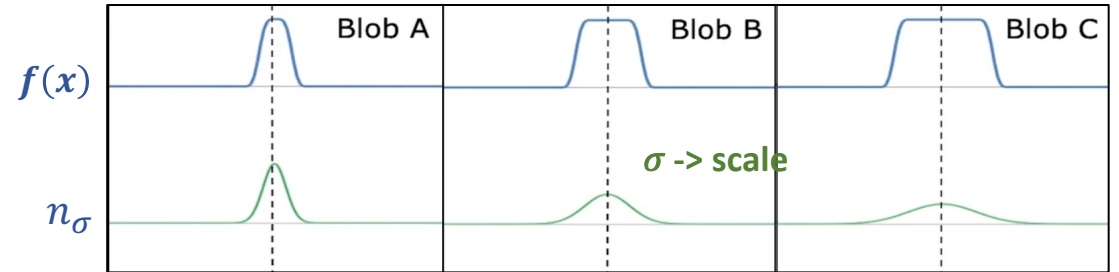
- The characteristic scale is proportional to the size of the blob.

- $$\frac{\text{size of blob A}}{\text{size of blob B}} = \frac{\sigma_A^*}{\sigma_B^*} \qquad \frac{\text{size of blob B}}{\text{size of blob C}} = \frac{\sigma_B^*}{\sigma_C^*}$$

XI. The SIFT Detector

Principle of blob detection

The algorithm for 1D signals



- 1) Given a 1D signal $f(x)$
- 2) Compute $\sigma^2 \frac{\partial^2 n_\sigma}{\partial x^2} * f(x)$ at k different scales $(\sigma_0, \sigma_1, \dots, \sigma_k)$
- 3) Find $(x^*, \sigma^*) = \arg \max_{(x, \sigma)} \left| \sigma^2 \frac{\partial^2 n_\sigma}{\partial x^2} * f(x) \right|$

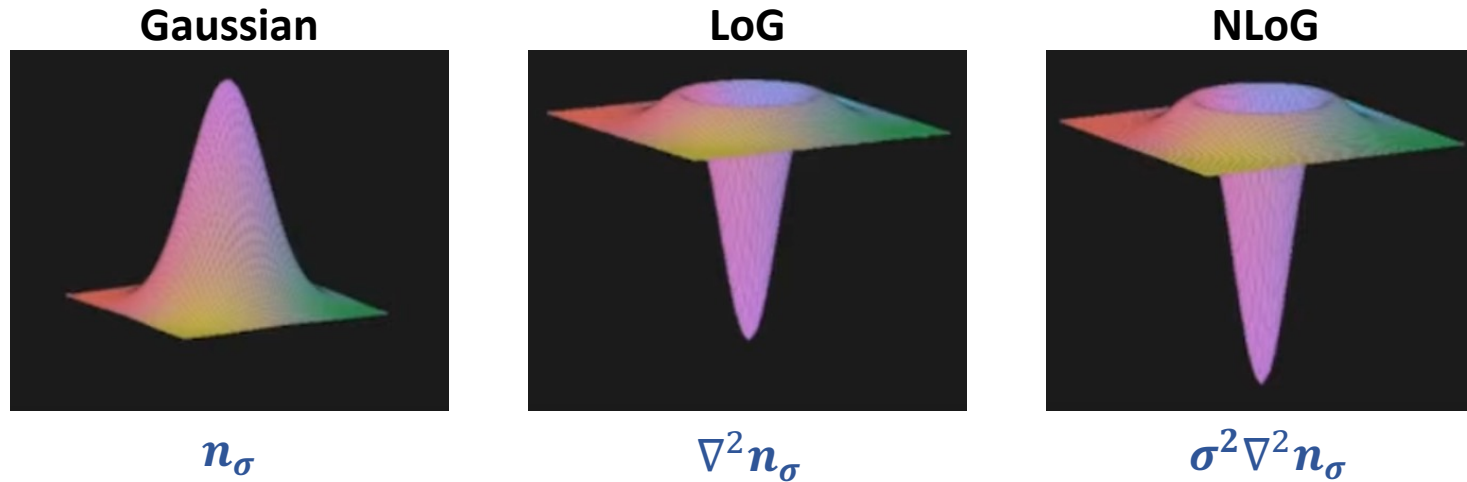
x^* : the blob position

σ^* : characteristic scale (blob size)

XI. The SIFT Detector

Principle of blob detection – 2D blob detection (in images)

The normalized Laplacian of Gaussian (NLoG) is used.

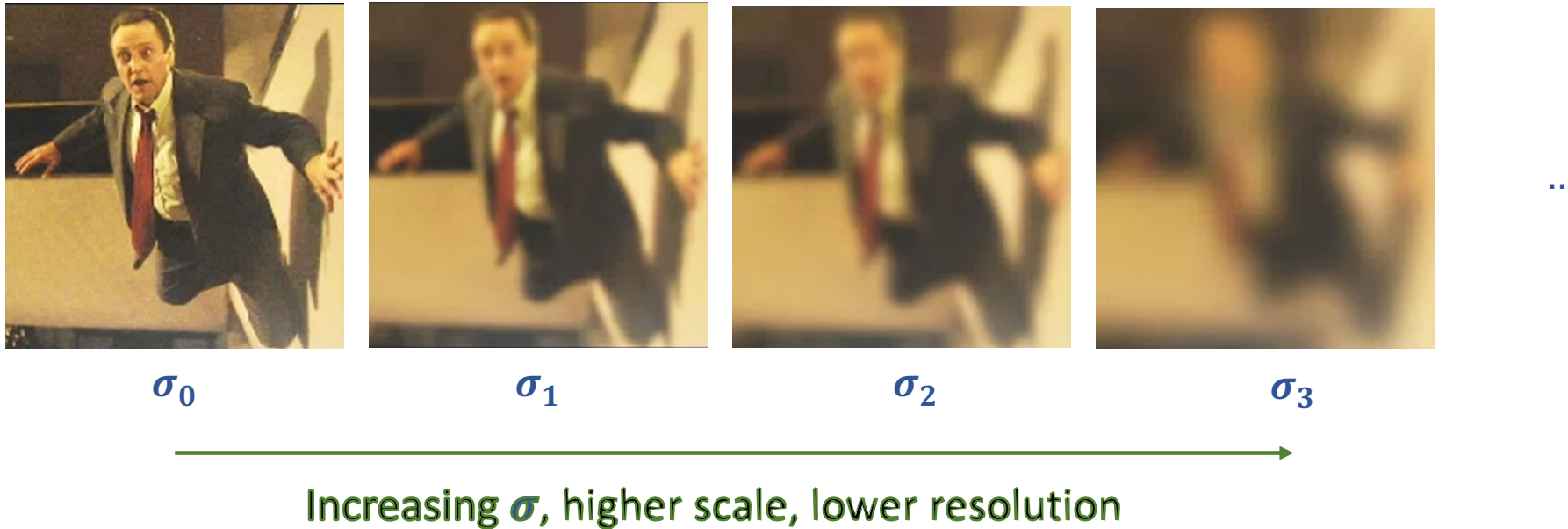


$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

Locations of the blobs are the maxima of the resulting 2D signal after applying NLoG...
at different scales, i.e. varying the value of σ

XI. The SIFT Detector

Principle of blob detection in images – An illustration

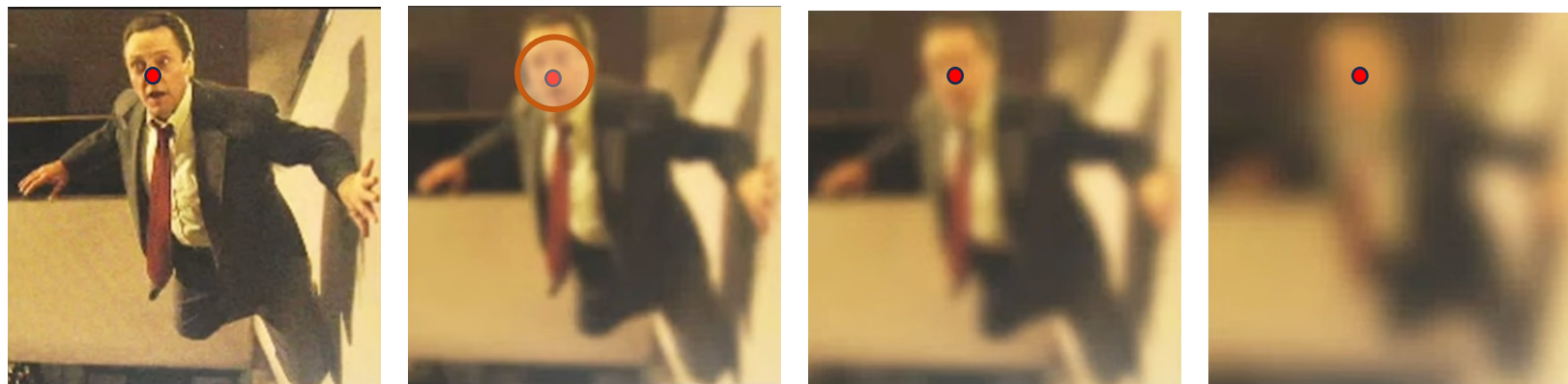


⇒ **Scale space:** A stack of images created by filtering images with different values of σ .

$$S(x, y, \sigma) = n(x, y, \sigma) * I(x, y)$$

XI. The SIFT Detector

Principle of blob detection in images – An illustration

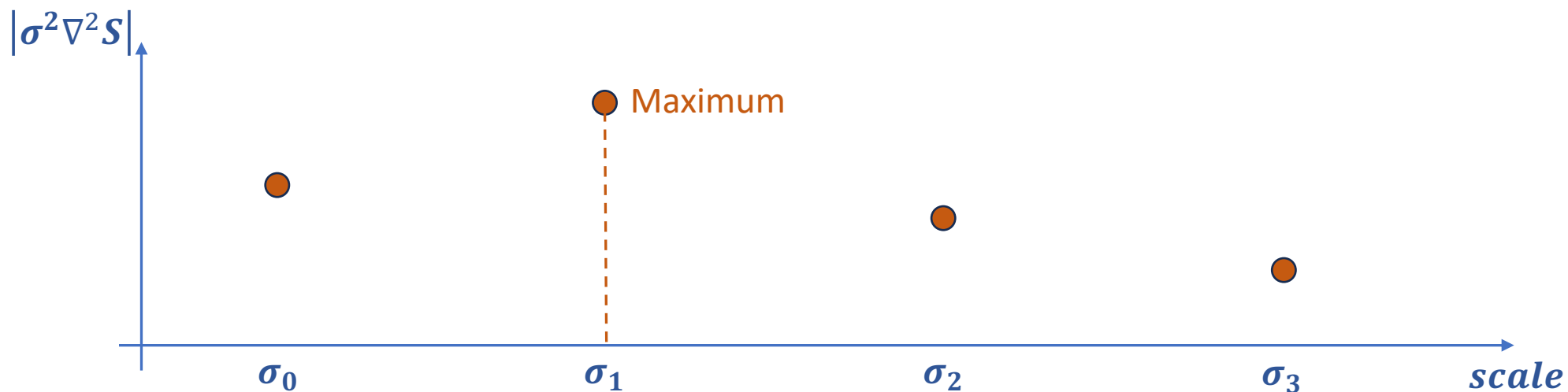


$S(x, y, \sigma_0)$

$S(x, y, \sigma_1)$

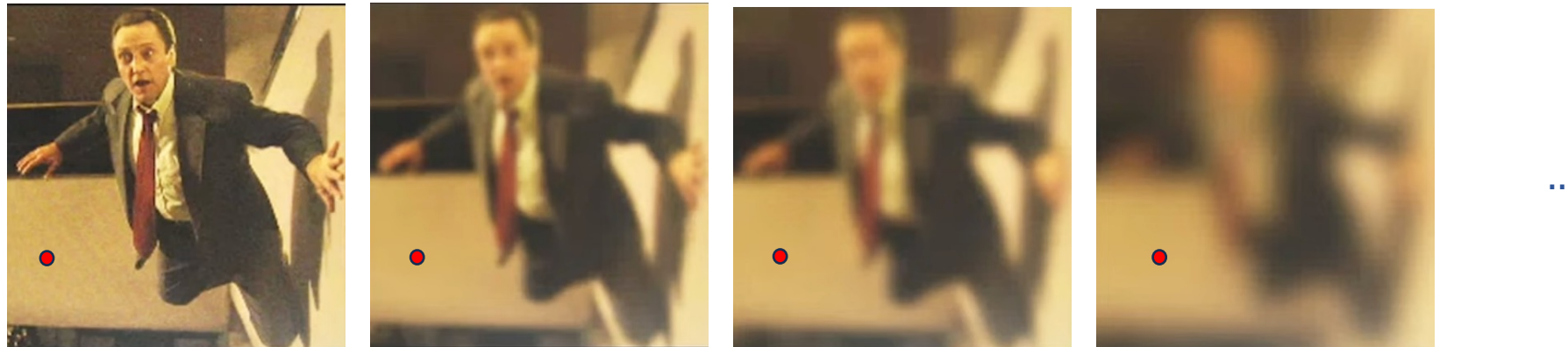
$S(x, y, \sigma_2)$

$S(x, y, \sigma_3)$



XI. The SIFT Detector

Principle of blob detection in images – An illustration

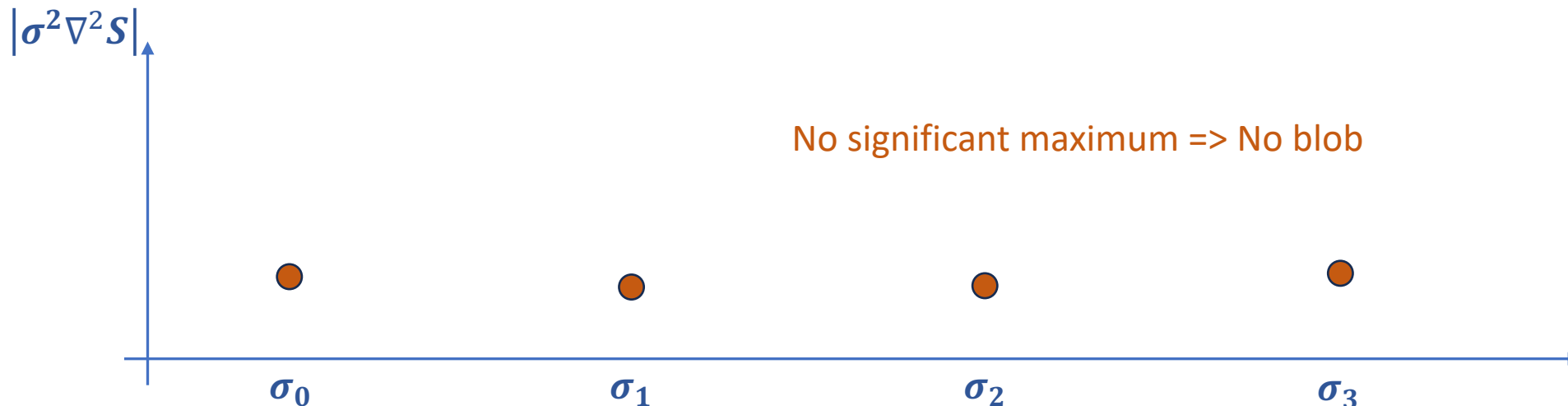


$S(x, y, \sigma_0)$

$S(x, y, \sigma_1)$

$S(x, y, \sigma_2)$

$S(x, y, \sigma_3)$



XI. The SIFT Detector

Principle of blob detection in an image

- 1) Given an image $I(x, y)$
- 2) Convolve the image using NLoG at different scales $(\sigma_0, \sigma_1, \dots, \sigma_k)$
- 3) Find $(x^*, y^*, \sigma^*) = \arg \max_{(x, y, \sigma)} |\sigma^2 \nabla^2 n_\sigma * I(x, y)|$

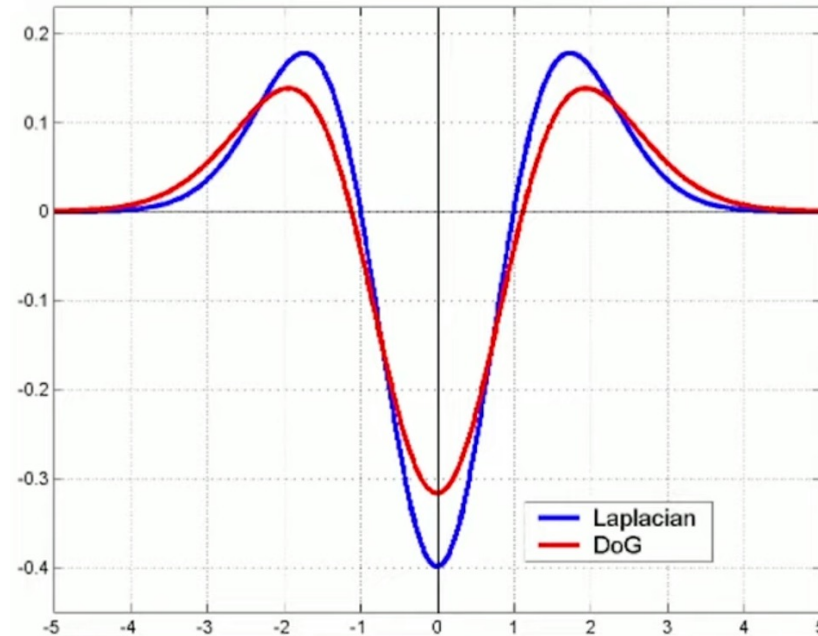
(x^*, y^*) : the blob position in the image I

σ^* : characteristic scale (blob size)

XI. The SIFT Detector

The SIFT blob detection

The difference of Gaussian (DoG) = $(n_{s\sigma} - n_s) \approx (s - 1)\sigma^2 \nabla^2 n_\sigma$

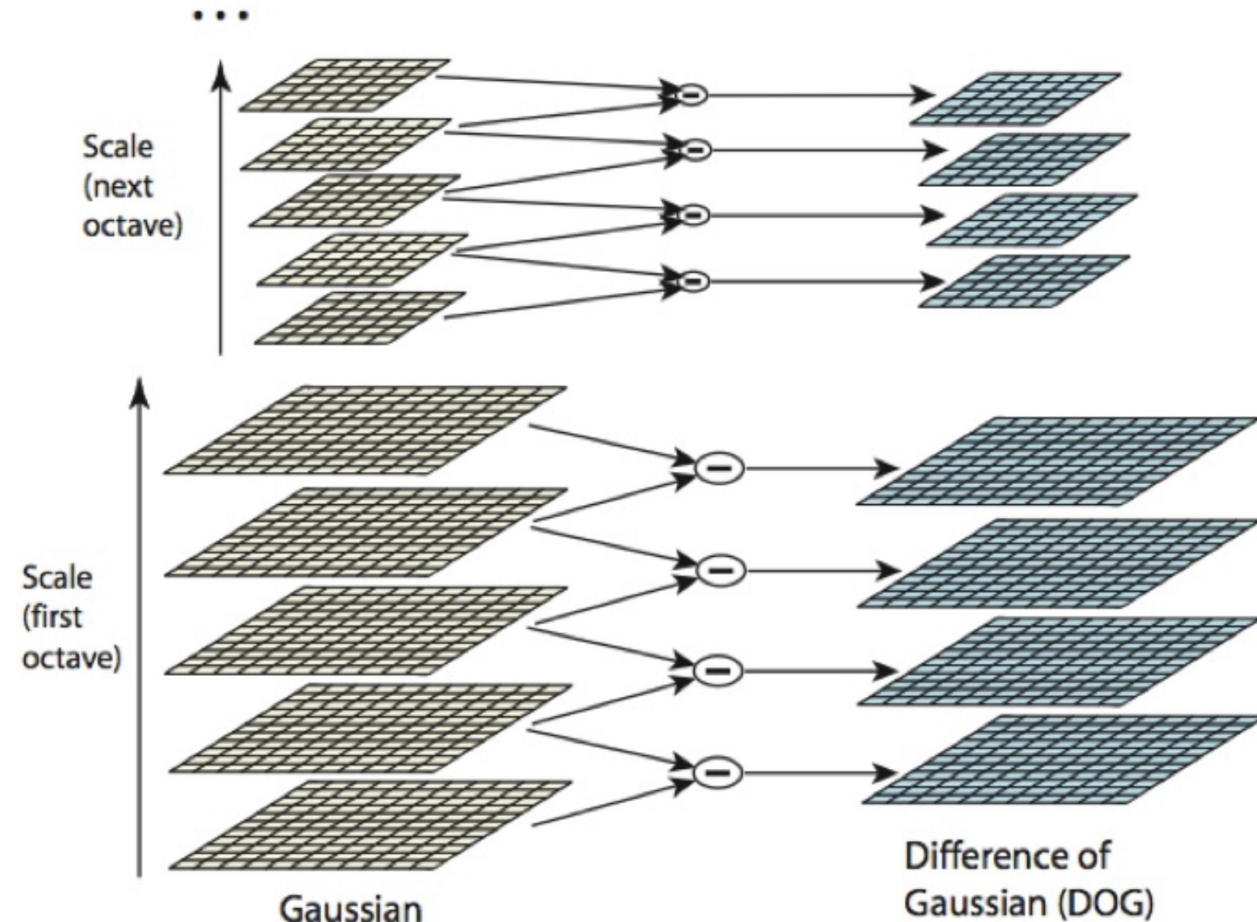


$$DoG \approx (s - 1)NLoG.$$

XI. The SIFT Detector

The SIFT blob detection

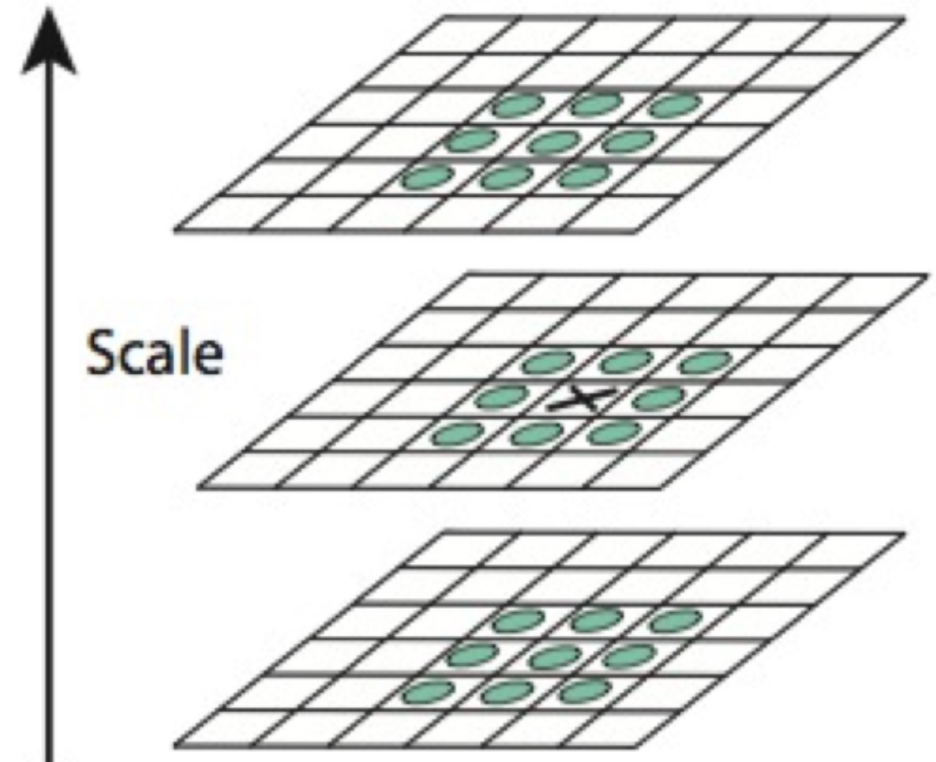
- Image is scaled several times (size is halved), each scale is called an octave.
- For each octave of scale space, the initial image is repeatedly convolved with Gaussians.
- Adjacent Gaussian images are subtracted to produce the DoG images.



XI. The SIFT Detector

The SIFT blob detection

- Once the DoG pyramids are built, the local-maxima are extracted considering both current-scale and adjacent scales.
- Local-maximum is checked against $9+8+9=26$ neighbours.
- Local-maxima $\Rightarrow$ Interest point candidates
- Apply a threshold to select the final SIFT interest points.



XI. The SIFT Detector

The SIFT blob detection

As you may have noticed, the scale is considered at two different steps:

1. Gaussian smoothing scale (σ)
2. Image resolution scale (octaves)

1. Gaussian Scale (σ)

- This refers to the **standard deviation** σ of the **Gaussian kernel** used to blur the image.
- The idea is to simulate observing the image at different "levels of detail" without changing the image size.
- For each fixed image resolution, the algorithm applies Gaussian blurs with increasing σ values.
- This forms the **scale-space** in the same **octave**.

XI. The SIFT Detector

The SIFT blob detection

As you may have noticed, the scale is considered at two different steps:

1. Gaussian smoothing scale (σ)
2. Image resolution scale (octaves)

2. Octaves (Image Resolution Scale)

- After a certain range of σ values, the image itself is **downsampled by a factor of 2** (width and height halved).
- This creates a **new octave**, i.e. a coarser version of the image.
- In each new octave, σ starts again from a base level (but represents a higher *absolute* scale due to the coarser image).

XI. The SIFT Detector

The SIFT blob detection

The SIFT algorithm constructs a **scale-space pyramid** made up of:

- Several **octaves** (image resolution levels),
- And, within each octave, several **blur levels** (controlled by increasing σ).

Each octave:

- Begins with a Gaussian-blurred image at some base σ .
- Applies further Gaussian blurs to get images at increasing σ values.
- Uses adjacent blurred images to compute **Difference of Gaussians (DoG)**.
- After several DoG levels, **the image is downsampled**, starting a new octave.

Gaussian σ controls smoothing.

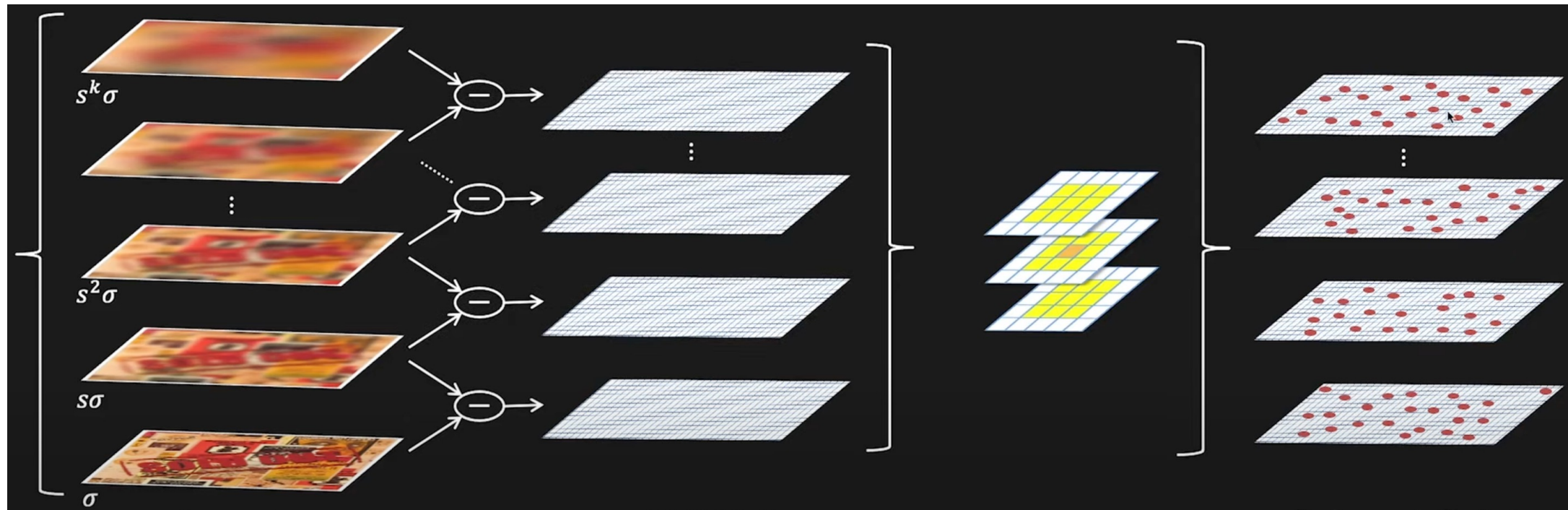
Octaves handle spatial resolution reduction.

⇒ Together, they form the full **scale-space** needed to detect **scale-invariant interest points**.

XI. The SIFT Detector

The SIFT blob detection

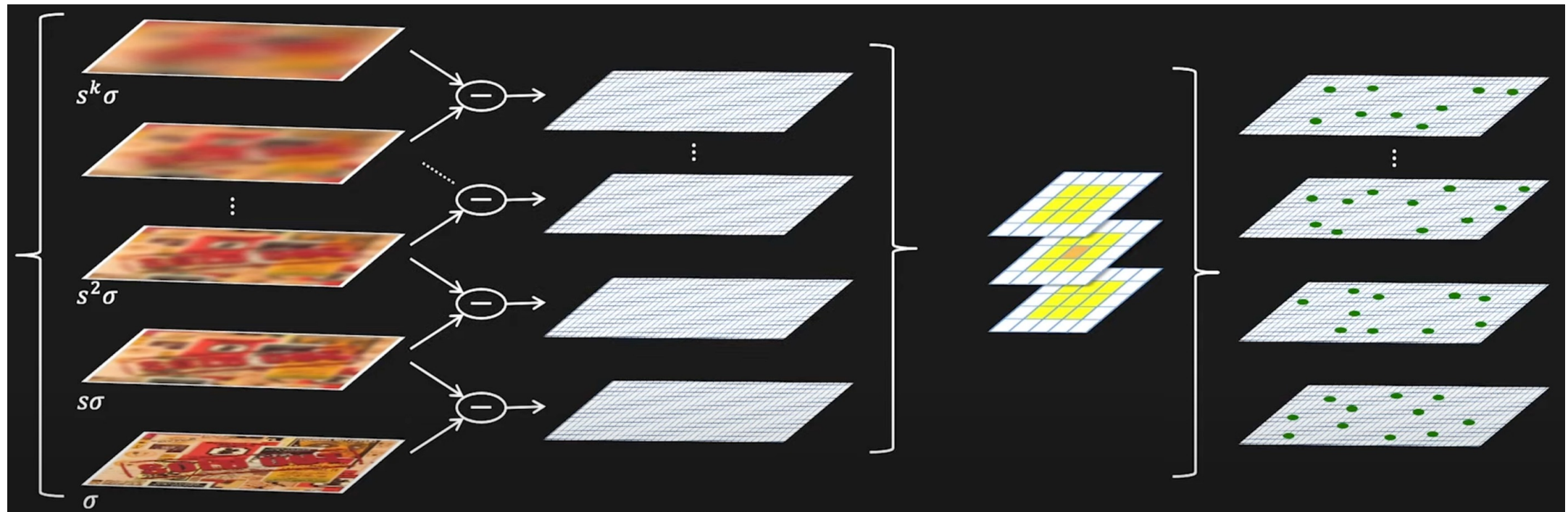
Another illustration for a specific **octave** (a fixed spatial resolution):



XI. The SIFT Detector

The SIFT blob detection

Another illustration for a specific **octave** (a fixed spatial resolution):

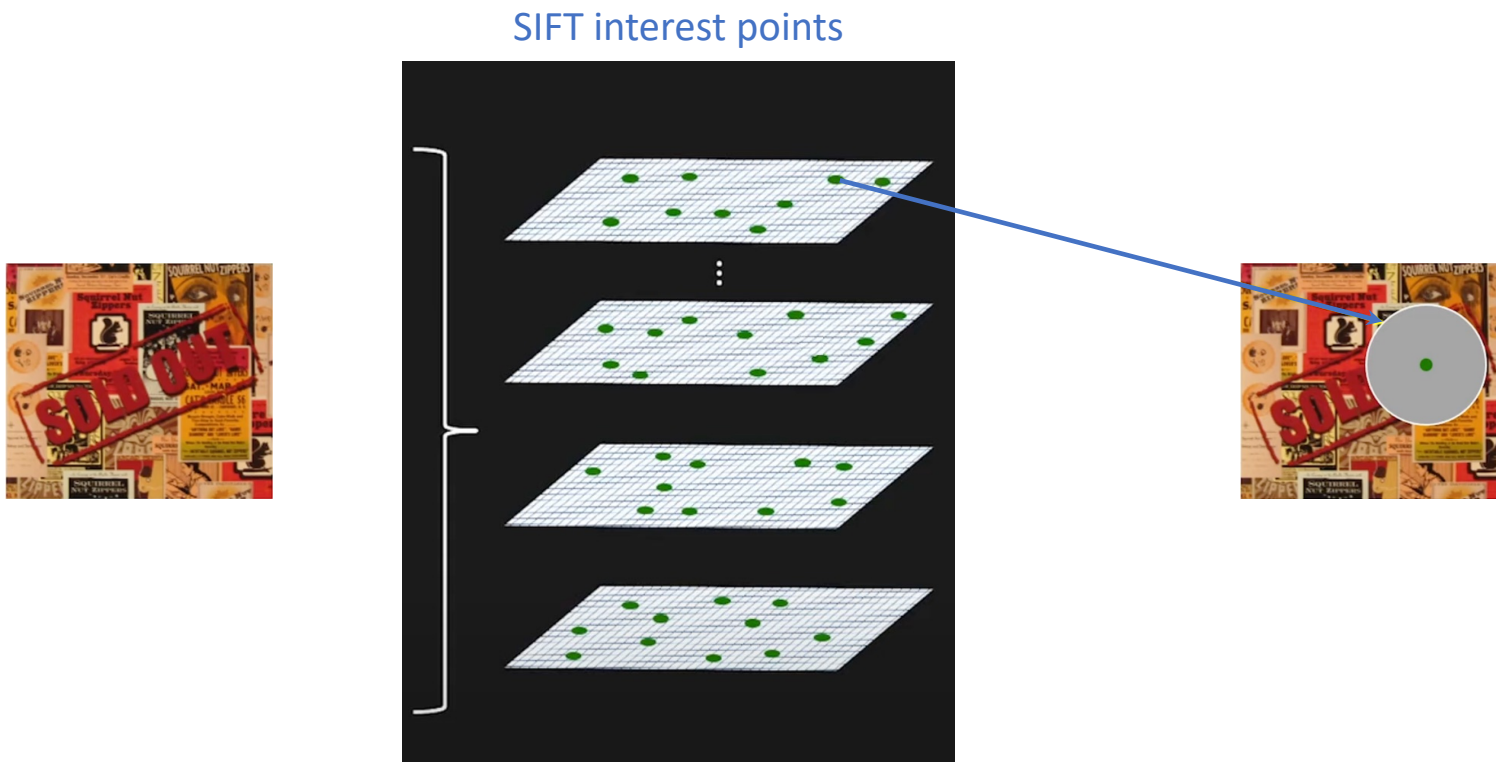


SIFT interest points

XI. The SIFT Detector

The SIFT blob detection

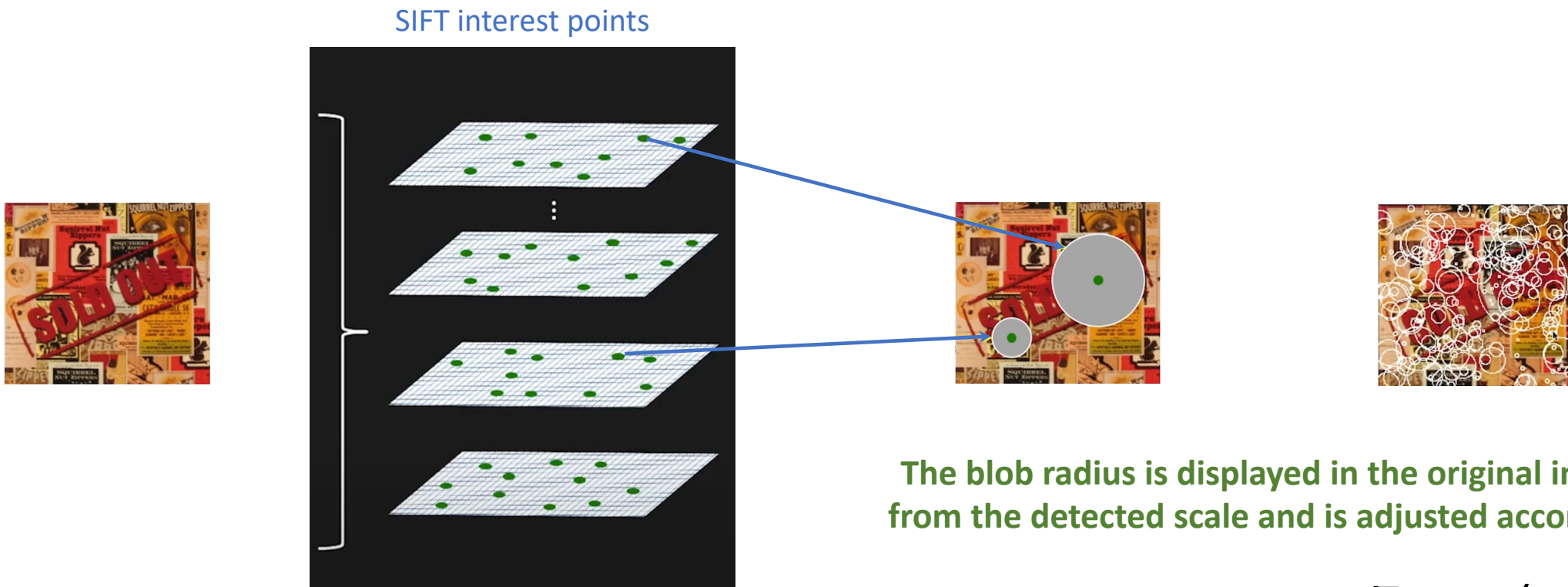
Another illustration for a specific **octave** (a fixed spatial resolution):



XI. The SIFT Detector

The SIFT blob detection

Another illustration for a specific **octave** (a fixed spatial resolution):



The blob radius is displayed in the original image, but it comes from the detected scale and is adjusted according to the octave.

$$r_{display} \approx \sqrt{2} \sigma \cdot 2^{octave}$$

XI. The SIFT Detector

The SIFT blob detection

Other examples



XI. The SIFT Detector

The SIFT blob detection: Dealing with variations in orientation

⇒ Assigning a principal orientation to each blob

The idea is to use the histogram of gradient directions

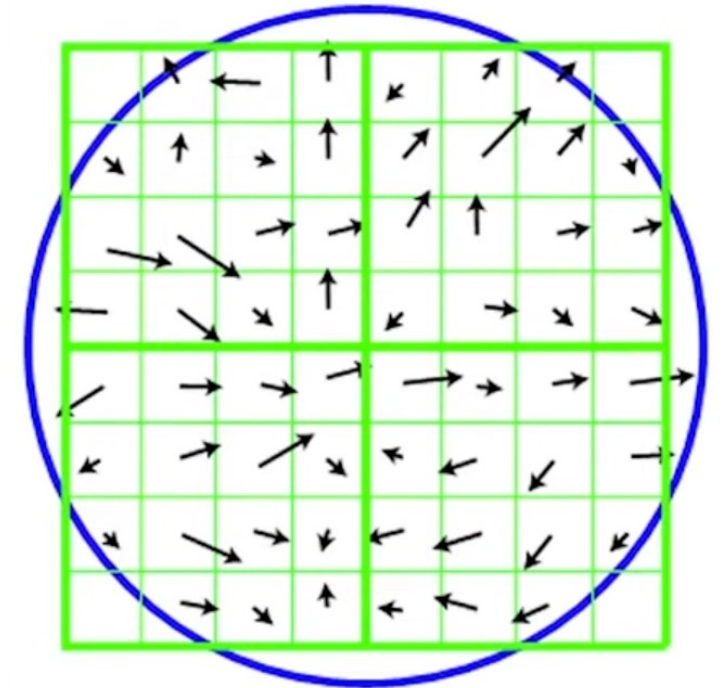
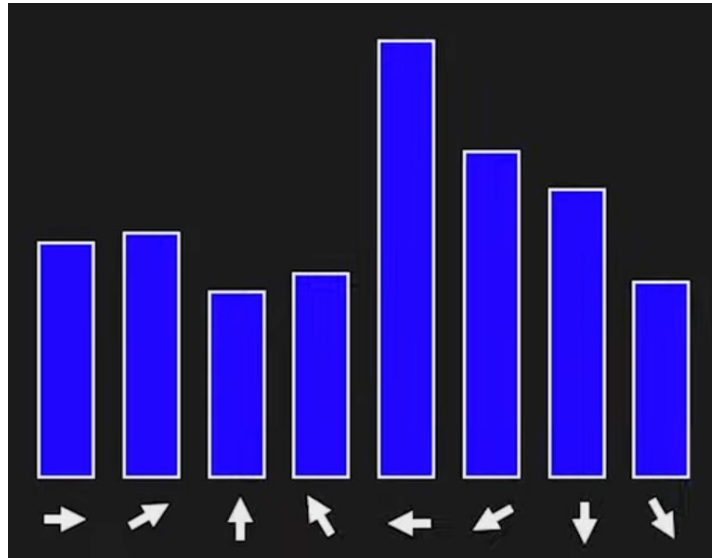


Image gradient direction

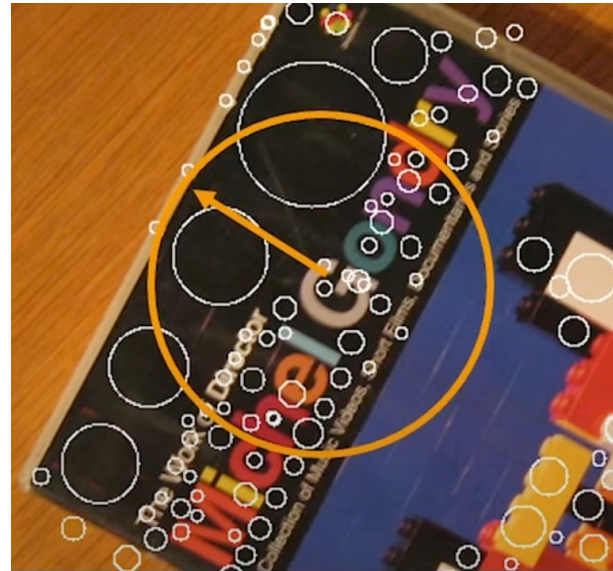
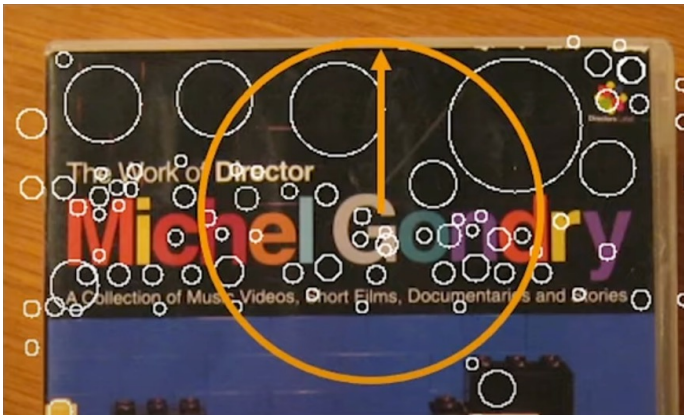
$$\theta = \tan^{-1} \left(\frac{\partial I}{\partial y} / \frac{\partial I}{\partial x} \right)$$

XI. The SIFT Detector

The SIFT blob detection

If we know the scale and the orientation, and we are able to match blobs:

⇒ We can undo the rotation and scale variations

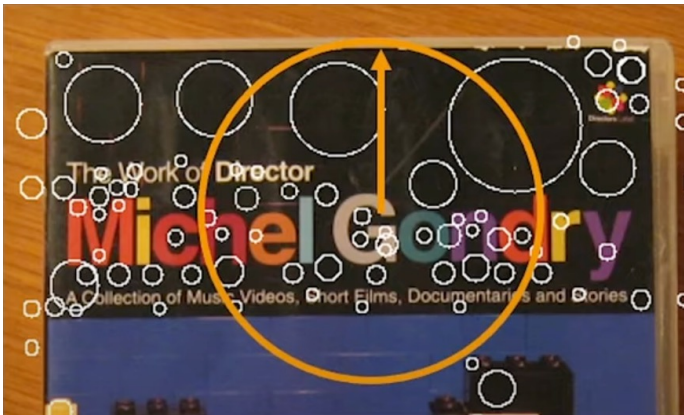


XI. The SIFT Detector

The SIFT blob detection

If we know the scale and the orientation, and we are able to match blobs:

⇒ We can undo the rotation and scale variations



XI. The SIFT Detector

The SIFT blob detection

If we know the scale and the orientation, and we are able to match blobs:

⇒ We can undo the rotation and scale variations



σ_1



σ_2

$$\text{Ratio of blob sizes} = \frac{\sigma_1}{\sigma_2}$$

XI. The SIFT Detector

The SIFT blob detection

If we know the scale and the orientation, and we are able to match blobs:

⇒ We can undo the rotation and scale variations



$$\text{Ratio of blob sizes} = \frac{\sigma_1}{\sigma_2}$$